Components for Optical Telecommunications
Long homework – Spring 2025
Due date: 16 May 2025

Topic 1

General description and rules:
You have until the due date to give back a max 10-page report on the topic described in the next page. It must include at least 15 references to scientific articles (no website allowed) formatted using the Optical Society of America template (Optics Express). The report must be written in LateX.
Remember to keep all original data calculated, evaluated, measured, they may be asked at some point. Same for the figures.

Each group has a separate topic. You can use all resources available. Reference articles can be downloaded from the UEF library website. Plagiarism will be checked. Consider that this is a real research work, which means that you must follow ethical rules.
Figures must be well done (no print screen allowed, except the one mentioned later). If you do not have any proper drawing software, I suggest PowerPoint.

The template for the report is the one provided by Optics Express. Follow their guidelines.

You can send me your report anytime by email only. I will give you some update, suggestions, comments until the due date. The report must be sent by email to me (matthieu.roussey@uef.fi) by the 16.05.2025 5pm sharp for evaluation and grading. No delay authorized.

The file you send me must be labeled as LH_2025_Group*N* (replace *N* by your group number!).

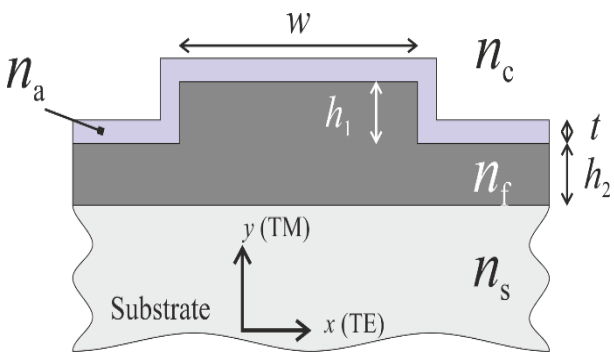
Grading:
This work counts for 30% of the total grade.

Criteria of evaluation	*Points*
Overall presentation	5
Writing	5
Bibliography	5
Introduction and conclusion	5
Presentation of the results	20
Discussion	20
Original thinking/innovation	10

Tool:
This work involves the use of three software available freely online:
OMS: 1D mode solver: https://www.computational-photonics.eu/oms.html
EIMS: 2D mode solver: https://www.computational-photonics.eu/eims.html
WMMS: Vectorial mode solver: https://www.siio.eu/wmms.html

Topic 3: Silicon rib waveguide with a conformal coating of titania

Let us consider a channel waveguide represented by its cross-section, see Fig. 1, called rib waveguide. The substrate is a fused silica wafer (SiO_2, $n_s = 1.444$ at $\lambda = 1550$ nm). The core of the waveguide is made of silicon (Si, $n_f = 3.48$ at $\lambda = 1550$ nm). The core is a thin film of Si partly patterned so that it forms a strip of width w and a height h_1, on top of the film. The parameters are depicted in Fig. 1. We deposit a thin layer of titanium dioxide (TiO_2, $n_a = 2.27$ at $\lambda = 1550$ nm) on top of this waveguide. The thickness of this conformal coating is $t = d$. The illumination wavelength is $\lambda = 1550$ nm.
Study the structure as a function of the different parameters and the refractive index of the surrounding material, and write a report presenting the results.



Parameters	Limit values
Width w	100 – 1000 nm
Height h_1	0 – 220 nm
Height h_2	0 – 220 nm
Height $h = h_1 + h_2$	220 nm
$t = d$	0 – 100 nm

Figure 1: Cross-section of the waveguide and its parameters

gh study of the electromagnetic field distribution in the t. I give below some hints on how to study the field and what to look for, but you must go beyond these few details.

Some key-things to study but you must go further:

- Vary the width w and the heights h_1 *and* h_2 of the waveguide core without a cladding to obtain the single mode behavior.
- Vary the thickness d.
- Study the evolution of the confinement factor Γ_a of the field inside the surrounding material as a function of the n_a and of the parameters of the waveguide.
- Evaluate the influence of polarization.
- Evaluate the validity of the effective index approximation on the effective index of the mode while passing from this two-dimensional simulation to the simplified one-dimensional case (% error).